

Homework 5

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (3-16). Fix a positive constant α , choose $x_1 > \sqrt{\alpha}$, and define

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right),$$

which determines $x_2, x_3, x_4, \dots$

(a) Prove that $\{x_n\}$ is monotonically decreasing and

$$\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}.$$

(b) Let $\varepsilon_n = x_n - \sqrt{\alpha}$. Prove that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}.$$

Hence, setting $\beta = 2\sqrt{\alpha}$, obtain

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n}, \quad n = 1, 2, 3, \dots$$

(c) This is a good method for computing square roots, because the recursion is simple and converges very fast. For example, if $\alpha = 3$ and $x_1 = 2$, prove that

$$\frac{\varepsilon_1}{\beta} < \frac{1}{10},$$

and therefore

$$\varepsilon_5 < 4 \cdot 10^{-16}, \quad \varepsilon_6 < 4 \cdot 10^{-32}.$$

Solution. (a) First, we will show (by induction) that $\sqrt{\alpha} < x_{n+1} < x_n$ for any positive integer n . It suffices to show that if $x_n > \alpha$, then $\alpha < x_{n+1} < x_n$. This is because

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) > \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha} \quad (\text{Notice that } x_n \neq \sqrt{\alpha}),$$

$$x_n - x_{n+1} = \frac{1}{2} \left(x_n - \frac{\alpha}{x_n} \right) = \frac{x_n^2 - \alpha}{2x_n} > 0.$$

Next, we will show that $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$. Since $\{x_n\}$ is bounded and monotone, it converges. Let $x = \lim_{n \rightarrow \infty} x_n$. Then

$$x = \frac{1}{2} \left(x + \frac{\alpha}{x} \right) \implies x^2 = \alpha \implies x = \sqrt{\alpha}.$$

(b) We have

$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} - 2\sqrt{\alpha} \right) = \frac{1}{2} \left(\sqrt{x_n} - \sqrt{\frac{\alpha}{x_n}} \right)^2 = \frac{\varepsilon_n^2}{2x_n}.$$

Since $x_n > \sqrt{\alpha}$, we have $\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$. Let $\beta = 2\sqrt{\alpha}$. By induction, we can show that $\varepsilon_{n+1} < \beta (\varepsilon_1/\beta)^{2^n}$ for any positive integer n . It suffices to show that if $\varepsilon_n < \beta (\varepsilon_1/\beta)^{2^{n-1}}$, then $\varepsilon_{n+1} < \beta (\varepsilon_1/\beta)^{2^n}$. This is because

$$\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}} < \frac{\beta^2}{2\sqrt{\alpha}} \left(\frac{\varepsilon_1}{\beta} \right)^{2^n} = \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n}.$$

(c) Take $\alpha = 3$ and $x_1 = 2$, we can calculate that

$$\varepsilon_1 = x_1 - \sqrt{\alpha} = 2 - \sqrt{3} \approx 0.268, \quad \beta = 2\sqrt{3} \approx 3.464 \implies \frac{\varepsilon_1}{\beta} \approx 0.077 < \frac{1}{10}.$$

Problem 2 (3-17). Fix $\alpha > 1$. Choose $x_1 > \sqrt{\alpha}$ and define

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}.$$

(a) Prove that

$$x_1 > x_3 > x_5 > \cdots.$$

(b) Prove that

$$x_2 < x_4 < x_6 < \cdots.$$

(c) Prove that

$$\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}.$$

(d) Compare the convergence rate of this method with that of the method in Exercise 16.

Solution. (a) and (b) From the recursion, we have

$$x_{n+2} - x_n = \frac{(1 + \alpha)x_n + 2\alpha}{2x_n + (1 + \alpha)} - x_n = -\frac{2(x_n^2 - \alpha)}{2x_n + (1 + \alpha)} \begin{cases} < 0, & \text{if } x_n > \sqrt{\alpha}, \\ > 0, & \text{if } x_n < \sqrt{\alpha}. \end{cases}$$

$$x_{n+1} - \sqrt{\alpha} = \frac{(\sqrt{\alpha} - 1)(\sqrt{\alpha} - x_n)}{1 + x_n} \begin{cases} < 0, & \text{if } x_n > \sqrt{\alpha}, \\ > 0, & \text{if } x_n < \sqrt{\alpha}. \end{cases}$$

Then the conclusions follow immediately from $x_1 > \sqrt{\alpha}$ and little effort on induction.

(c) Since $\{x_{2n-1}\}$ is bounded and monotone, it converges. Let $x = \lim_{n \rightarrow \infty} x_{2n-1}$. Then

$$x = \frac{(1 + \alpha)x + 2\alpha}{2x + (1 + \alpha)} \implies x = \pm\sqrt{\alpha}.$$

Since all the terms are positive, we have $x = \sqrt{\alpha}$.

Similarly, we can show that $\lim_{n \rightarrow \infty} x_{2n} = \sqrt{\alpha}$. Therefore, $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$.

(d) Let $\varepsilon_n = x_n - \sqrt{\alpha}$. We have

$$\varepsilon_{n+1} = \frac{(\sqrt{\alpha} - 1)(\sqrt{\alpha} - x_n)}{1 + x_n} = \frac{(\sqrt{\alpha} - 1)\varepsilon_n}{1 + x_n}.$$

Then the convergence rate is linear, i.e. $\varepsilon_n \sim \left(\frac{\alpha-1}{\alpha+1}\right)^n$, which is much slower than the method in Exercise 16.

Problem 3 (3-20). Let $\{p_n\}$ be a Cauchy sequence in a metric space X , and suppose that a subsequence $\{p_{n_i}\}$ converges to a point $p \in X$. Prove that the whole sequence $\{p_n\}$ converges to p .

Proof. For any positive real number ε , there exists integer N_1 and N_2 such that $d(p_{n_i}, p) < \varepsilon/2$ for all $n_i \geq N_1$ and $d(p_m, p_n) < \varepsilon/2$ for all $m, n \geq N_2$.

Take $N = \max\{N_1, N_2\}$. For any $n > N$, select $n_i > n$. Then by the definition of N , we have

$$d(p_n, p) \leq d(p_n, p_{n_i}) + d(p_{n_i}, p) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence, $\{p_n\}$ converges to p . □

Problem 4 (3-21). Prove the following theorem, analogous to Theorem 3.10(b): if, in a complete metric space X , each E_n in the sequence $\{E_n\}$ is a nonempty bounded closed set, $E_n \supset E_{n+1}$, and

$$\lim_{n \rightarrow \infty} \text{diam}(E_n) = 0,$$

then

$$\bigcap_{n=1}^{\infty} E_n$$

consists of exactly one point.

Proof. First, we will show that $\bigcap_{n=1}^{\infty} E_n$ is nonempty. For each positive integer n , select $p_n \in E_n$. Since E_1 is bounded and $p_n \in E_n \subset E_1$ for any $n \leq 1$, The sequence $\{p_n\}$ is bounded. Therefore, $\{p_n\}$ has a convergent subsequence, say $\{p_{n_i}\}$. Denote $\lim_{i \rightarrow \infty} p_{n_i} = p$.

For any $n \in \mathbb{N}_+$, since $\{p_{n_i} : n_i \geq n\} \subset E_n$, E_n is closed, and $p_{n_i} \rightarrow p$, we have $p \in E_n$. Thus, $p \in \bigcap_{n=1}^{\infty} E_n$, and the intersection is nonempty.

Next, we will show that $\bigcap_{n=1}^{\infty} E_n$ consists of exactly one point. Suppose that there are two distinct points $p, q \in \bigcap_{n=1}^{\infty} E_n$. Then for any positive integer n , since $p, q \in E_n$, we have $d(p, q) \leq \text{diam}(E_n)$. Letting $n \rightarrow \infty$, we have $d(p, q) = 0$, which is a contradiction. Hence, the intersection consists of exactly one point. □